Dylan Dsouza

San Diego, CA

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Education

University of California San Diego San Diego, CA
Bachelor of Science in Data Science – GPA: 3.92 (Provost Honors) *Expected March 2027*

Experience

Data Science Instructor Apr 2026 – Present
San Diego Supercomputer Center *San Diego, CA*

- Designed and taught an introductory data science curriculum for high school students through the UCSD Junior Research Academy, covering Python (pandas), EDA, data visualization, AI literacy, and data ethics via live coding demos in Google Colab.
- Guided 6 students with no prior programming experience through independent data projects from dataset selection to final Lightning Talk presentations, building end-to-end data analysis and research communication skills.

Research Data Analyst May 2025 – Present
Scripps Institution of Oceanography *San Diego, CA*

- Automated 11 years of coral reef data processing (60% cut in manual processing time), building a Python ETL pipeline that standardized 40,000+ benthic annotations into clean datasets for downstream analysis across 15 reef sites.
- Conducted EDA and time series analyses across 5 experimental groups and 16 survey time points to investigate treatment effects on reef dynamics, identifying temporal trends and ecological patterns to inform restoration strategies.

Market Research Intern Jul 2023 – Aug 2023
Bennett Coleman & Co. Ltd. (The Times Group) *Mumbai, India*

- Built a multiclass NLP classifier achieving 85% accuracy across 5 market sectors, processing 56,000+ newspaper headlines using TF-IDF and BERT embeddings to automate market sector categorization at scale.
- Produced an internal research brief on data infrastructure modernization – covering data lakes, CDPs, and CRM optimization – to support a 5-person team's evaluation of editorial and business workflow improvements.

Leadership & Activities

AI Research & Development Analyst Oct 2025 – Present
Computer Science & Engineering Society *San Diego, CA*

- Implemented a hybrid Neo4j retrieval pipeline fusing MiniLM vector search and BM25 via Reciprocal Rank Fusion, and built the React/Vite frontend for an agentic AI platform modeling professional decision-making styles.
- Constructed a knowledge graph pipeline extracting LLM-generated concepts from research documents, mapping entity relationships via contextual proximity scoring and Girvan-Newman community detection across 8+ entity categories.

Machine Learning Researcher Oct 2025 – Present
Engineers for Exploration *San Diego, CA*

- Engineered a Google Earth Engine pipeline to stage 69-band Sentinel-2 and satellite embedding training tiles across Madagascar and Mozambique, gating downloads to mangrove regions using ESA WorldCover land-cover classification.
- Integrated 7 geospatial datasets to curate a structured training dataset for pixel-level mangrove mapping, processing 21,000+ GeoTIFF image-mask pairs across 4 land-cover classes for downstream model training.

Projects

ENSOcast: Interactive Climate Analytics Dashboard | *Conference Project (SDUTC 2025)* View Project | GitHub

- Engineered a Streamlit climate dashboard on 40+ years of NOAA data; benchmarked Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble models achieving 82% accuracy in 3-class ENSO phase prediction.

The Palmyra Restoration Experiment | *Conference Project (MURALS 2026)* View Project | GitHub

- Modeled 10 years of benthic community recovery across 5 coral transplantation treatments at Palmyra Atoll, quantifying how restoration design shapes long-term benthic community trajectories.

America on Fire: Mapping U.S. Wildfire Distribution | *Course Project (Data Visualization)* View Project | GitHub

- Visualized 2024 U.S. wildfire distribution through an interactive D3.js map using NASA MODIS satellite data, surfacing spatial burn patterns, seasonal trends, and unexpected regional hotspots.

NextMove: Predicting Next Workout Activity | *Course Project (Recommender Systems)* View Project | GitHub

- Predicted next workout type from 20,000+ Endomondo records, engineering temporal and physiological features to achieve 87% accuracy with Random Forest over 3 heuristic baselines.

DegreeDelta: Quantifying the Income Advantage of College Degrees | *Personal Project* View Project | GitHub

- Quantified the college income premium across all 50 U.S. states using weighted ACS PUMS microdata and state-level aggregation, revealing geographic variation in the financial return to higher education.

Skills

Languages: Python, SQL (PostgreSQL), R, Java, HTML, CSS, JavaScript
Libraries/Frameworks: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PySpark, Matplotlib, Plotly, NLTK, LangChain, AWS, Git
Core Competencies: Data Cleaning/Visualization, EDA, ETL Pipelines, A/B Testing, Financial Modeling, Time Series Analysis, NLP